Project Report

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Computer Graphics

Section: N
Group No: 01

Project Report on:

Interactive Animated Village and Camping Scene using OpenGL

Submitted to:

Mahamodul Hasan Mahadi
Lecturer, Department of Computer Science
American International University-Bangladesh

Submitted By:

Name	ID	CONTRIBUTION (%)
Md. Saiful Kabir	23-53874-3	25%
Monami Maymuna	23-54885-3	25%
Arpita Sen Arpa	23-54186-3	17%
Morioum Meem	23-54138-3	17%
Salam Bin Faruki	23-54059-3	16%

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1. Introduction

Computer Graphics is one of the most important fields of computer science for creating visual environments, animations, simulations, and interactive graphical systems. Real-world objects and environments can be represented through shapes, colors, transformations, lighting effects, and animations using graphics programming. OpenGL provides efficient rendering functions and real-time animation support, so it is one of the most widely used graphics libraries for implementing graphical applications .

The main purpose of this project was to develop an "Interactive Animated Village & Camping Scene" using OpenGL and C++. The project represents a realistic village environment beside a river where different natural elements and animated objects work together to create a dynamic graphical simulation. The entire scene contains mountains, forests, trees, houses, rivers, boats, clouds, birds, flowers, roads, camping tents, campfire, airplane animation, weather effects, and environmental transitions.

The project was designed with different interactive features so that users can control multiple environmental conditions using keyboard input. In this project we can see day, sunset and night view with rain and thunderstorm effects. Boats, clouds, plane, fire and birds animation also added in this project to create a realistic atmosphere.

In this project we used OpenGL primitives such as polygons, triangles, quadrilaterals, lines, and triangle fans to create graphical objects. We also followed transformation techniques such as translation, scaling, and mirroring to create movement and animation effects.

This project provided practical experience in OpenGL rendering, animation systems, event handling, environmental simulation, and real-time graphics programming.

2. Objective

- To develop an interactive 2D animated village and camping scene using OpenGL and C++ with environmental elements such as mountains, rivers, trees, houses, and camping areas.

- To implement animations for moving boats, birds, clouds, planes, fire effects, rain, and lightning to create a dynamic environment.

- To create smooth transitions between day, sunset, and night using changing sky colors and object visibility.

- To apply important Computer Graphics concepts such as rendering, transformation, animation, scaling, translation, and object layering through practical implementation.

- To develop user interaction systems using keyboard controls so that users can dynamically control weather conditions, environmental modes, boat movement, and other animations during runtime.

3. Tools and Technologies Used

3.1 Programming Language

The programming language used for the project was C++. C++ was selected because it provides high execution speed and works efficiently with OpenGL graphics libraries.

3.2 Graphics API

OpenGL was used as the main graphics API to deliver all objects and animations in the project.

3.3 FreeGLUT Library

The FreeGLUT library was used to create the application window, handle rendering events, manage keyboard input, and control timer-based animation systems.

3.4 IDE and Compiler

Code::Blocks IDE with the MinGW compiler was used to write, compile, and run the project on the Windows operating system.

3.5 Mathematical Functions

Mathematical calculations such as trigonometric operations using sine and cosine functions were used to create circular objects and smooth animation effects.

3.6 Multimedia Library

The Windows multimedia library was also used to implement rain sound effects using the PlaySound() function.

4. System Description

4.1. Environment and Sky

The environment system controls the overall appearance of the scene. The sky changes dynamically depending on the current environmental state. During the day, the sky color shows like bright blue, while during the sunset warm orange shades are visible at the sky. At night view, the sky becomes dark, and the moon becomes visible. Clouds continuously move across the sky using translation animation. Different sky colors and object colors are applied automatically during day, sunset, and night modes to create realistic environmental transitions.

4.2 Mountain and Tree

The project contains multiple mountains and trees to create a natural village environment. Mountains were created using layered triangles with different shades to create depth and realism. Different styles of trees such as pine trees, circular leaf trees, and forest trees were designed using polygons and circular shapes. Grass fields and flowers were added throughout the scene to improve visual beauty and create a realistic village atmosphere.

4.3 River and Boat

A river was designed in the middle portion of the scene using polygon rendering and gradient color combinations. The river creates a natural separation between different environmental layers. Two different boats were implemented in the project. Both boats move continuously across the river using translation animation. Users can control boat movement and speed using keyboard input.

4.4 Camping and Fire

The camping area consists of a tent, wooden logs, and an animated campfire. The tent was created using polygon and triangle shapes with multiple color layers. The campfire animation becomes visible mainly at night. Different flame layers with scaling animation were used to create realistic fire flickering effects. The fire continuously changes its size and flame tip position to simulate natural flame movement. This camping area increases the realism and attractiveness of the village environment.

4.5 Weather and Rain

The project includes a complete weather simulation system consisting of rain and lightning effects. Rainfall animation was implemented using repeated line rendering techniques. Rain particles continuously fall from the top of the screen during rain mode. Lightning effects were created using line strip primitives and blinking animation logic. Random lightning flashes appear across the sky during storm mode. Rain sound effects were also added using the Windows multimedia library to make the storm environment more realistic.

4.6 Bird and Airplane Animation

Bird animation was implemented using line primitives and wing movement logic. Birds continuously move across the sky while their wings flap dynamically using animation variables. An airplane animation system was also implemented in the project. The airplane moves across the sky using translation and scaling transformations. Scaling creates a perspective effect that makes the airplane appear distant while moving upward. Both birds and airplanes make the environment more dynamic and visually attractive.

4.7 User Interaction

A complete keyboard interaction system was implemented to allow users to control different environmental and animation features during runtime. Users can switch between day, sunset, and night modes. Rain and lightning effects can also be turned on or off using keyboard controls. Boat movement, boat speed, and airplane movement can also be controlled. This interaction system makes the project more dynamic and user-friendly.

5. Implementation Details

5.1 Initialization

The initialization part was used to set up the OpenGL environment before starting the project. The background color was set using glClearColor(), and the 2D coordinate system was created using gluOrtho2D(). The ortho is set from 0 to 500 on both the X-axis and Y-axis. Also, different global variables were initialized to control animations and environment effects. Object movement and scene changes were controlled by variables like dayState, boatSpeed, cloudX, birdWing, rainY, and planeX. Here initial positions of sun, moon, clouds, boats and other animated objects were also set.

5.2 Display Function

The display() function was used for drawing the entire scene on the screen. At first, the screen was cleared using glClear(). Then all objects were drawn one by one. Different transformation functions such as glTranslatef() and glScalef() were used to move and animate objects properly.

5.3 Timer Function

The update() timer function controlled all animations in the project. The function was called every 16 milliseconds using glutTimerFunc() to keep the animations smooth. Boat movement, cloud movement, bird movement, airplane movement, fire animation, rain effect, and lightning blinking were updated inside this function. The timer function also used for the change between day, sunset, and night scenes. glutPostRedisplay() continuously refreshed the screen so that all animations could run smoothly in real time.

5.4 Drawing Techniques

OpenGL primitives such as GL_POLYGON and GL_QUADS were mainly used for drawing houses, rivers, roads, tents, etc. GL_LINES and GL_LINE_STRIP were used for rain and lightning effects. Circular objects such as the sun, moon, clouds, flowers, and tree leaves were created using GL_TRIANGLE_FAN and trigonometric calculations. Different color combinations were also used to create realistic environmental views.

6. Algorithm (Step-by-Step)

6.1 Circle Drawing Algorithm

Circles were used to draw objects like the sun, moon, clouds, flowers, and some tree leaves. The circles were created using GL_TRIANGLE_FAN along with trigonometric calculations.

6.2 2D Transformation

2D transformation was used for moving and positioning different objects in the scene. The project mainly used translation and scaling transformations for object movement and animation.

6.3 Timer-based Animation

The update() timer function controlled all animations in the project. The function was called every 16 milliseconds using glutTimerFunc() to keep the animations smooth and continuous.

6.4 Color Interpolation

Color interpolation was used for the sunset effect and environmental transitions. Different RGB values were changed gradually to create smooth transitions between day, sunset, and night views.

6.5 Gradient Coloring

Gradient coloring was used in the sky and river. Different colors were applied to the top and bottom vertices of the same shape. OpenGL blended these colors automatically and created a smooth gradient effect. This technique made the sky and river look more realistic.

6.6 Line Drawing Technique

Line drawing was used for rain, lightning, bird wings, and some object details. Rain was created using many short line segments falling from the top of the screen. Lightning was created using connected line segments with GL_LINE_STRIP.

6.7 Keyboard Interaction

Keyboard interaction was used to control different parts of the project. The keyboard() function handles user input.

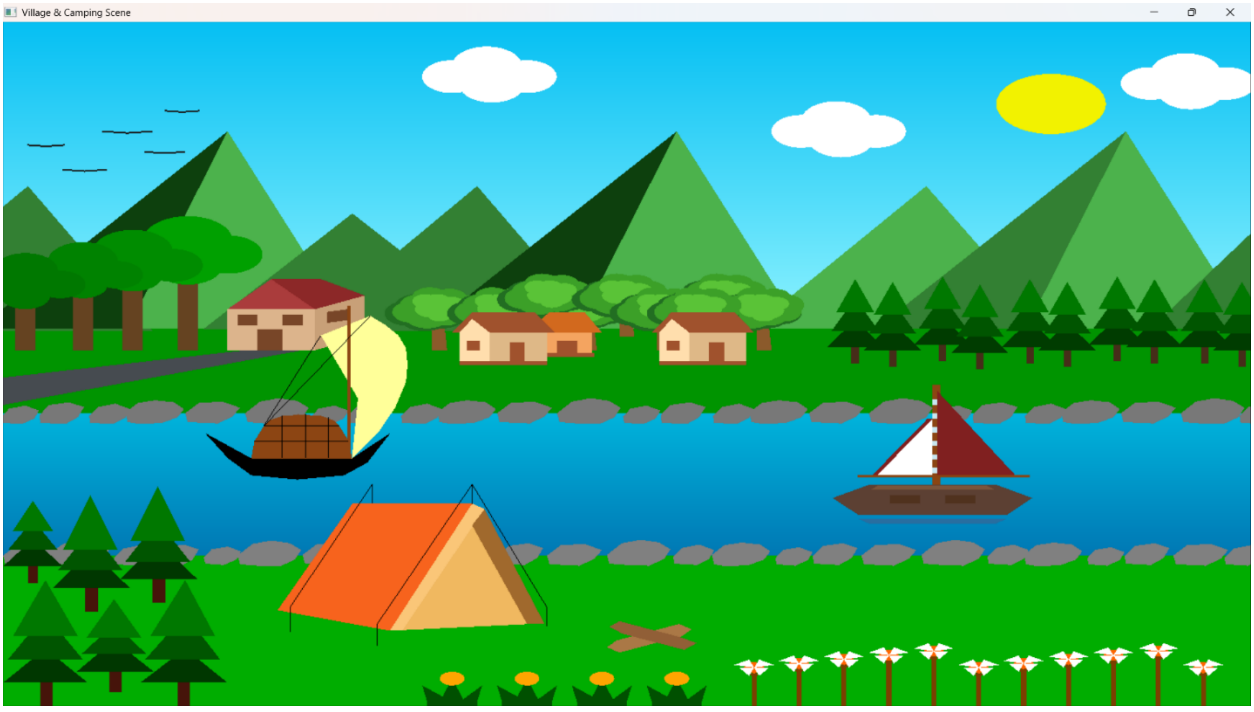
6.8 Layering / Object Ordering

The objects were drawn in a proper order so that the scene looked correct. Background objects such as sky, sun, moon, clouds, mountains, houses, river, and ground were drawn at first. Finally, foreground objects such as boats, tent, fire, rain, and lightning were drawn. This helped maintain proper visual depth in the scene.

7. Output Description

Full project is demonstrated here step by step:

7.1 Day Mode (Default)



When the project starts, the default environment appears in daytime mode. The sky becomes bright blue and the sun appears in the upper portion of the scene. Mountains, trees, houses, flowers, river, and camping area become clearly visible with bright environmental colors. Clouds move continuously across the sky and birds fly dynamically using animation. Two boats move across the river from opposite directions. The camping area contains a tent, wooden logs, flowers, and decorative environmental elements that improve the visual appearance of the village & camping environment.

Users can switch to daytime mode by pressing the **'d'** key.

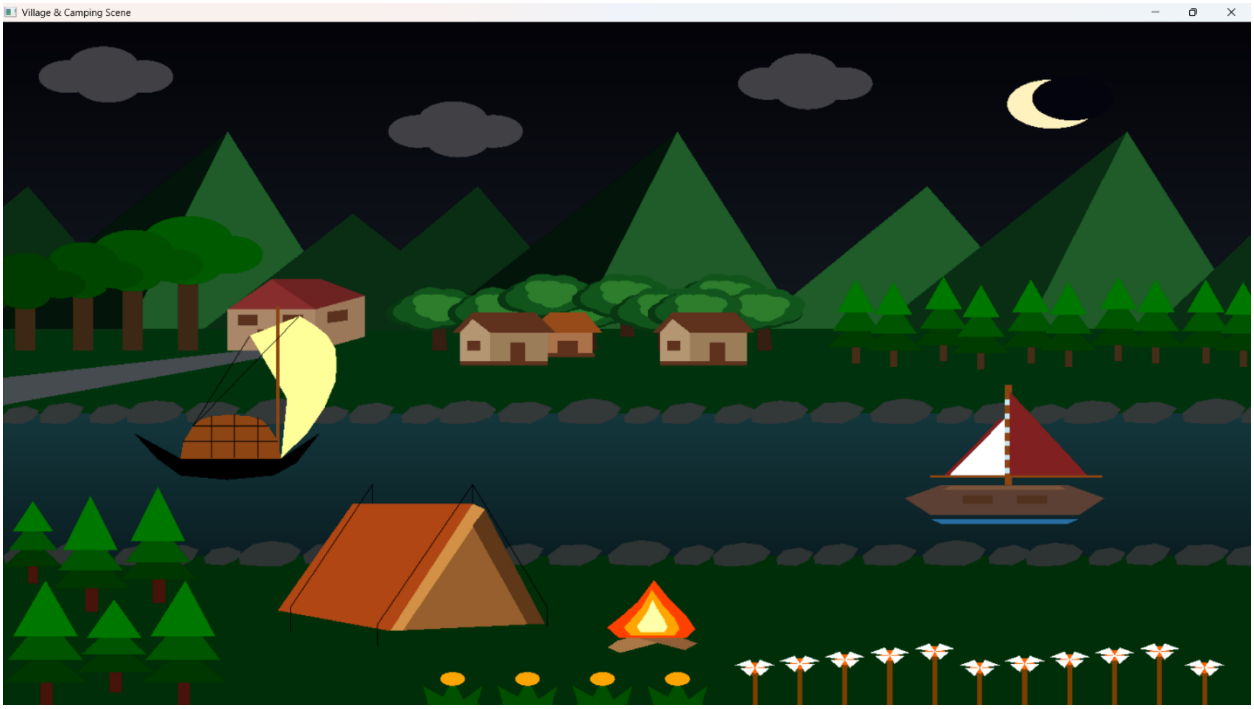
7.2 Sunset Mode



The project also contains a sunset environment system. During sunset mode, the sky color gradually changes from blue to orange and purple shades using color interpolation techniques. The environmental colors become warmer and darker compared to daytime. Clouds receive sunset color effects to create a realistic evening atmosphere. Birds continue flying and boats continue moving normally during this mode. The sunset environment creates a visually attractive transition between day and night.

Users can activate sunset mode by pressing the **'m'** key.

7.3 Night Mode



During night mode, the sky becomes dark and the moon appears in the upper sky area. Environmental brightness decreases and darker color combinations are applied to mountains, trees, river, and houses. The campfire near the tent becomes more visible at night. Clouds continue moving slowly across the sky while boats continue river movement. The night environment creates a peaceful and realistic nighttime village scene.

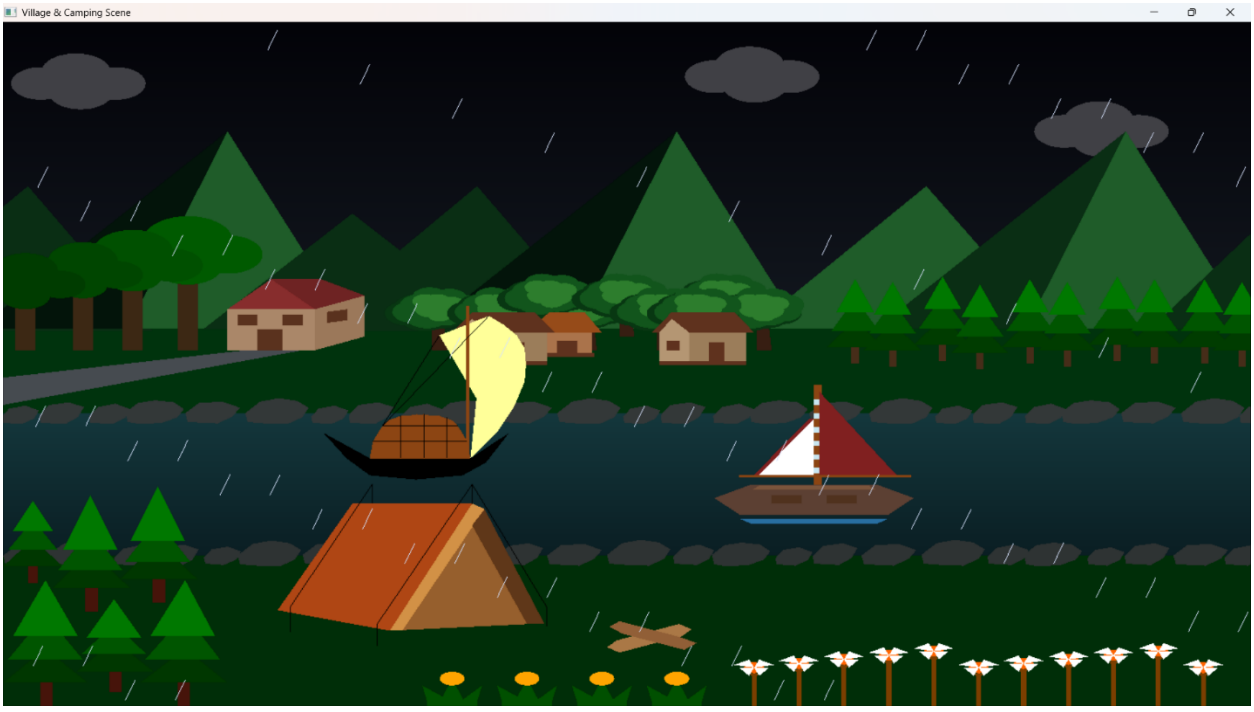
Users can switch to night mode by pressing the **'n'** key.

7.4 Rainy Weather

The project contains a rain simulation system which can be activated using keyboard controls. During rain mode, multiple rain particles fall continuously from the top of the screen using animated line rendering. Rain can appear during both daytime and nighttime environments. The sky becomes darker during rain mode to create a storm-like atmosphere. The river, mountains, houses, and camping area remain visible while rain falls throughout the scene. The weather system increases the realism and dynamic appearance of the project. Rain sound effects were also added using the Windows multimedia library to improve the overall user experience.

Users can turn rain ON by pressing the **'r'** key and turn rain OFF using the **'1'** key.


Village & Camping Scene


Village & Camping Scene

7.5 Thunderstorm Effect



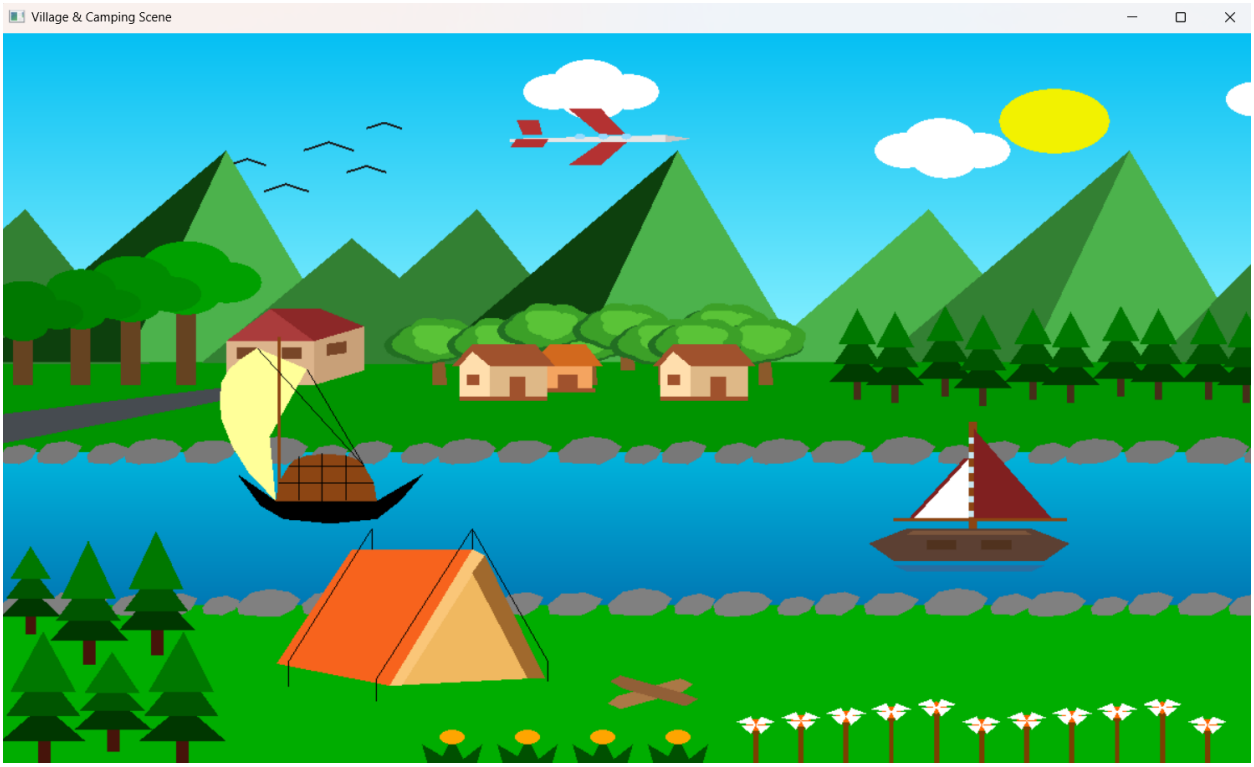


A thunderstorm effect was also implemented in the project using animated lightning flashes. White lightning bolts appear randomly in the sky using line strip drawing techniques. The lightning effect works together with the rain system to create a realistic storm environment.

During nighttime storms, the dark sky combined with rain and lightning creates a more dramatic visual effect.

Users can turn lightning ON using the **'t'** key and turn it OFF using the **'2'** key.

7.6 Plane and Bird Animation



Bird animation was implemented using wing movement logic and translation techniques. Multiple birds continuously fly across the sky to create natural environmental movement. An airplane animation system was also added to the project. The airplane moves smoothly across the sky and creates additional environmental activity during daytime scenes. Both airplane and bird movement help make the environment feel more alive and interactive.

Users can start airplane movement from the left side using the **'p'** key and from the right side using the **'o'** key.

7.7 Boat Animation

Two different boats were implemented in the river section of the project. Both boats move continuously in opposite directions using translation animation. One boat contains a curved traditional sail design while the other contains a triangular sail structure. The movement

speed of the boats can be controlled through keyboard interaction. The moving boats make the river environment more dynamic and visually appealing.

Boat 1 Controls:

- **'a'** → Start boat movement
- **'x'** → Stop boat movement
- **'f'** → Increase boat speed
- **'s'** → Decrease boat speed
- **'z'** → Reset normal speed

Boat 2 Controls:

- **'g'** → Start boat movement
- **'h'** → Stop boat movement
- **'j'** → Increase boat speed
- **'k'** → Decrease boat speed
- **'l'** → Reset normal speed

8. Result

The project was successfully implemented using OpenGL and C++. All animation systems, environmental transitions, and interaction systems worked properly. The project successfully displayed realistic village scenery with animated environmental effects. Boat movement, bird movement, cloud movement, airplane animation, rain effects, lightning systems, and campfire animation were implemented successfully. The user interaction system also worked correctly through keyboard input. Environmental transitions between daytime, sunset, and nighttime were smooth and visually attractive. The final output successfully demonstrated practical implementation of computer graphics concepts and OpenGL animation systems.

9. Conclusion

This project provided valuable practical experience in OpenGL graphics programming and real-time animation systems. Through this project, different graphical concepts such as

rendering, transformations, animation loops, environmental simulation, and event handling were learned and implemented successfully. The project successfully demonstrated how a complete animated environment can be created using OpenGL primitives and animation techniques. Different natural objects, weather systems, moving objects, and interactive controls were integrated together to create a realistic graphical simulation. The implementation of environmental transitions, weather systems, and dynamic animations improved understanding of computer graphics and real-time rendering systems. Overall, the project achieved its objectives successfully and created a complete interactive village and camping scene using OpenGL and C++.

10. Limitation & Future Improvements

Limitations:

Only 2D Graphics Environment Used

The entire project was developed using 2D OpenGL graphics. Although the scene contains multiple animated objects and environmental effects, realistic 3D depth, camera perspective, and object rotation systems were not implemented. Because of this, the visual environment looks simpler compared to modern 3D graphical simulation systems.

Limited User Interaction

User interaction in the project is mainly controlled through keyboard input. Users can control weather systems, environmental modes, boats, and airplane movement, but there are no mouse interaction systems, camera movement controls, or object selection features.

Limited Sound Effects

Only rain sound effects were added using the Windows multimedia library. Other environmental sounds such as thunder sound, river water sound, wind sound, campfire sound, and bird sound were not included in the current implementation.

Future Improvements:

Future 3D Environment Development

In the future, the entire village and camping scene can be converted into a 3D environment using modern OpenGL techniques. This would create more realistic mountains, rivers, houses, trees, and environmental depth.

Improved Lighting and Shadow System

Advanced lighting systems such as dynamic shadows, reflection effects, texture mapping, and realistic illumination models can be added to improve the visual quality of the project.

More Interactive Features

Additional interaction systems such as mouse control, zooming, camera movement, and object interaction can be implemented to make the project more interactive and user-friendly.

Additional Environmental Effects

More realistic environmental features such as snowfall, fog, moving animals, water wave simulation, and seasonal changes can be added in future versions to make the environment more dynamic and visually attractive.

Enhanced Audio System

Future versions of the project may include multiple environmental sound effects such as rain, thunder, birds, river flow, and campfire sound to create a more immersive user experience.

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